

Deep Gaussian Scale Mixture Prior for Image Reconstruction

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Abstract—Image reconstruction from partial observations has attracted increasing attention. Conventional image reconstruction methods with hand-crafted priors often fail to recover fine image details due to the poor representation capability of the hand-crafted priors. Deep learning methods attack this problem by directly learning mapping functions between the observations and the targeted images can achieve much better results. However, most powerful deep networks lack transparency and are nontrivial to design heuristically. This paper proposes a novel image reconstruction method based on the Maximum a Posterior (MAP) estimation framework using learned Gaussian Scale Mixture (GSM) prior. Unlike existing unfolding methods that only estimate the image means (i.e., the denoising prior) but neglected the variances, we propose characterizing images by the GSM models with learned means and variances through a deep network. Furthermore, to learn the long-range dependencies of images, we develop an enhanced variant based on the Swin Transformer for learning GSM models. All parameters of the MAP estimator and the deep network are jointly optimized through end-to-end training. Extensive simulation and real data experimental results on spectral compressive imaging and image super-resolution demonstrate that the proposed method outperforms existing state-of-the-art methods.

Index Terms—Gaussian scale mixture prior, maximum a posterior, spectral compressive imaging, single image super-resolution, swin transformer.

I. INTRODUCTION

IN REAL-WORLD imaging systems, observed images suffer from various degradations in spatial or spectral dimensions, resulting in nonideal performance at downstream tasks such as object recognition [1], [2], detection [3], [4], [5], and tracking [6]. These negative effects can be alleviated by computational image reconstruction methods [7], [8], [9], [10], [11],

[12], [13], [14], estimating high-quality images from their partial observations (e.g., the compressive image or low-resolution image). Mathematically, many image reconstruction problems can be represented as $y = Ax + n$, where y and x denote the observation and the original high-quality image, respectively, A denotes the measurement matrix of the imaging system, and n denotes the additive noise. According to different settings of the measurement matrix A , image reconstruction can be classified into different sub-tasks (e.g., spectral compressive imaging [9], [10], image super-resolution [15], [16], image denoising [7], [8], or image deblurring [15], [17], [18]).

For example, for spectral compressive imaging (SCI), the coded aperture snapshot spectral imaging (CASSI) system [19], [20], [21], [22], [23] utilizes a physical mask and a disperser to modulate different wavelength signals and mixes all modulated signals to generate a single 2D compressive image at video rate, thus A can be represented as a sampling matrix implemented by the coded aperture and disperser. For single image super-resolution (SISR), the observation is low-resolution due to long imaging distance or the limited resolution of the imaging sensor, thus A can be represented as a spatial downsampling matrix consisting of a blurring operator and a downsampling operator.

Generally, existing image reconstruction methods can be roughly classified into two categories, *model*-based and *learning*-based. Most existing model-based methods adopted hand-crafted priors to solve this ill-posed inverse problem. The commonly used hand-crafted priors include the total variation (TV) prior [24], [25], the sparsity-base prior with respect to some learned dictionaries [22], [26], [27], and the nonlocal self-similarity prior [8], [9], [28], [29]. However, due to the poor representation capability of these hand-crafted priors and the ineffective manual adjustment of the parameter setting, model-based methods achieved limited successes in recovering fine image details. In addition, model-based methods are time-consuming.

Inspired by the successes of the deep convolutional neural networks (DCNNs) for various vision tasks, deep learning-based image reconstruction methods [10], [14], [20], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42] have advanced rapidly. On one hand, various DCNN-based image reconstruction methods have also been proposed, which directly learn the mapping functions between the observations and the high-quality images via Generative Adversarial Network (GAN) [30], [37], deep residual networks [31], [34], [36], [43], densely-connected networks [33], attention-based

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networks [20], [34], [36], [44] and so on. Most recently, image reconstruction has benefited from the advances of the novel self-attention block, i.e., Transformer [45], such as SwinIR [46] and MST [47]. Although these DCNN-based and Transformer-based methods achieved state-of-the-art results, it is non-trivial to design a powerful DCNN or Transformer network heuristically. On the other hand, to design the network more transparently, many unfolding networks [11], [13], [32], [35], [38], [39], [48] have been developed based on iterative image reconstruction algorithms. However, most of them using simple convolutional layers, e.g., [38] and [39], are far from enough to learn comprehensive and sufficient image priors. In addition, all these unfolding methods only estimate the image mean (i.e., the denoising prior) but neglected the variance (i.e., the uncertainty in mean estimation). In other words, the variances of all pixels are same and fixed. This assumption is not consistent with the fact that the variances of different areas have very different variances.

Bearing the above concerns in mind, this paper proposes an image reconstruction method, named DGSM, with learned Gaussian Scale Mixture (GSM) priors. The contributions of this paper are as follows:

- The proposed method DGSM is an unfolding network using a novel deep GSM prior. In contrast with existing unfolding methods that only estimate the image means but neglect the variances, we propose characterizing images by the GSM models and estimate the means and the variances (i.e., the scale prior) of the GSM models through a DCNN.
- Considering that learning the means and the variances of the GSM models by DCNN ignores the long-range dependencies of the images, we develop an enhanced variant based on the Swin Transformer [49] for learning long-range image dependencies.
- The image reconstruction problem is formulated as a Maximum a Posteriori (MAP) estimation problem with the learned GSM models. All parameters in the MAP estimator are jointly optimized in an end-to-end manner.
- Extensive experimental results on both synthetic and real datasets show that the proposed method outperforms existing state-of-the-art SCI and SISR methods.

This paper is based on our conference paper [10]¹ and adds significant improvements in the following aspects:

- Considering that images often include rich repetitive structures, many existing image reconstruction methods [8], [35], [39] have benefited from learning long-range dependencies for recovering more image information. However, the proposed method in the previous version that learns the GSM priors (means and variances) by a DCNN without learning long-range dependencies achieved limited performance. In the present version, we develop an enhanced variant that learns long-range dependencies of pixels by stacking a series of Swin Transformer layers. Unlike the proposed method in the previous CVPR version, which learns a 3D filter for each pixel to estimate the local means

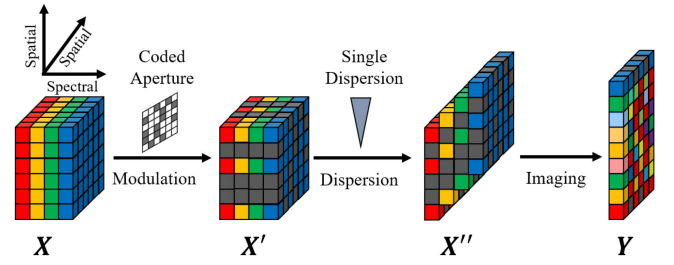


Fig. 1. The imaging schematic of the coded aperture snapshot spectral imaging (CASSI) system.

of the GSM models, the GSM priors in the present version are directly learned by the Swin Transformer network. Specifically, the proposed method in the present version outperforms the one in our previous conference version by 3.92 dB for SCI and 0.65 dB, 1.47 dB, and 0.20 dB with scales $\times 2$, $\times 3$, and $\times 4$ on the Urban100 testset for SISR, respectively.

- We have extended the proposed DGSM method to the single image super-resolution (SISR) task. Extensive experimental results on both synthetic and real datasets show that the proposed method can achieve comparable results with existing state-of-the-art SISR methods.

The rest of this paper is organized as follows. Section II briefly reviews existing image reconstruction works and Vision Transformer for image restoration. Section III-B proposes an image reconstruction method with learned GSM priors. Sections IV and V report our experimental results and compare them against existing state-of-the-art methods. We draw conclusions in Section VI.

II. RELATED WORK

We briefly review image reconstruction methods in Section II-A and the recently proposed Vision Transformer methods for image restoration in Section II-B.

A. Image Reconstruction

1) *Hyperspectral Image Reconstruction Methods for Spectral Compressive Imaging: The CASSI Observation Model.* Fig. 1 shows the imaging schematic of the coded aperture snapshot spectral imaging (CASSI) system. Let $X \in \mathbb{R}^{H \times W \times L}$ denote the original 3D hyperspectral image (HSI). As shown in Fig. 1, the HSI X is first modulated spatially by a coded aperture (i.e., a physical mask). Then, the modulated image $X' \in \mathbb{R}^{H \times W \times L}$ is dispersed by the following dispersive prism. The dispersive prism can cause the modulated image X' to be shifted in the spatial dimension at an interval of k pixels, and other empty positions are filled with 0. Since the modulated image X' has L channels, it will be shifted by $k \cdot (L - 1)$ pixels in the spatial dimension in total. Therefore, the dispersed image X'' has a size of $H \times (W + k \cdot (L - 1)) \times L$. For convenience, we let $k = 1$, thus the size of X'' is $H \times (W + L - 1) \times L$. At last, the dispersed image is compressed to a 2D compressive image $Y \in H \times (W + L - 1)$ and captured by a 2D imaging sensor.

¹The project website of the preliminary conference version is available at <https://see.xidian.edu.cn/faculty/wsdong/Projects/DGSM-SCI.htm>.

The modulated image $\mathbf{X}'$, the l^{th} wavelength of the dispersed image $\mathbf{X}''$, and the 2D compressive image $\mathbf{Y}$ can be represented respectively as

$$\begin{aligned}\mathbf{X}'_l &= \mathbf{C} \odot \mathbf{X}_l, \\ \mathbf{X}''(r, c, l) &= \mathbf{X}'(r, c + d_l, l), \\ \mathbf{Y} &= \sum_{l=1}^L \mathbf{X}''_l,\end{aligned}\quad (1)$$

where $\mathbf{C} \in \mathbb{R}^{H \times W}$ denotes the physical mask, $\odot$ denotes the element-wise product, d_l denotes the shifted distance of the l^{th} channel, $1 \leq r \leq H$, $1 \leq c \leq W$ and $1 \leq l \leq L$. Furthermore, considering the measurement noise $\mathbf{n} \in \mathbb{R}^M$, the matrix-vector form of the CASSI observation model can be formulated as

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}, \quad (2)$$

where $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{y} \in \mathbb{R}^M$ denote the vectorized form of $\mathbf{X}$ and $\mathbf{Y}$ respectively, $N = HWL$ and $M = H(W + L - 1)$, and $\mathbf{A} \in \mathbb{R}^{M \times N}$ denotes the measurement matrix of the CASSI system, implemented by the physical mask and disperser prism.

Conventional Model-Based HSI Reconstruction Methods. Existing model-based methods used various hand-crafted priors to help reconstruct the original HSI from the compressive image. In [27] and [50], gradient projection algorithms and dictionary learning methods were proposed to solve the ill-posed inverse problems with sparse regularizers, respectively. In [25], [26], [51], the total variation (TV) prior, as another commonly used hand-crafted prior, was used to suppress noise and artifacts. De-SCI [9] exploited the nonlocal self-similarity and the low-rank property of HSIs to reconstruct the original HSI. The major drawback of these model-based methods is that the hand-crafted priors show poor representation capability. In addition, these model-based methods are time-consuming.

Learning-Based HSI Reconstruction Methods. Due to the powerful learning ability of deep neural networks, learning-based methods achieved much better results than model-based methods by directly learning mapping functions between the observations and the targeted images. [52] first obtained the initial estimates of the HSIs by [51] and further refined the initial estimates by a DCNN. Like [52], λ -net [37] also adopted a two-stage network to reconstruct the HSIs. Different from [52], λ -net [37] used a Generative Adversarial Network (GAN) with self-attention to reconstruct an initial HSI and a refinement network to improve image quality. To take advantage of the spatial-spectral correlation of HSIs, TSA-Net [20] proposed a novel spatial-spectral self-attention module and achieved better results than λ -net [37]. In [44], the high-resolution dual-domain learning network (HDNet) was proposed, which includes spatial-spectral domain learning and frequency domain learning for reconstructing high-resolution fine-grained details. The first Transformer-based HSI reconstruction method MST [47] introduced Transformer to learn spectral-wise self-attention and developed a novel mask-guided mechanism to modulate HSI feature representations, leading to state-of-the-art performance. Different from the above learning-based methods that designed the DCNN and Transformer network heuristically,

deep unfolding methods [23], [38], [39], [53] unfolded HSI reconstruction algorithms with denoising priors into DCNNs and Transformer network. In DSSP [38], the spatial-spectral prior has been proposed but not been fully exploited, leading to limited success. Since the nonlocal self-similarity of HSIs has been exploited, DNU [39] achieved further improvements on DSSP [38]. GAP-CCoT [23] proposed a novel convolution and contextual Transformer (CCoT) block and integrate it into the generalized alternating projection (GAP) algorithm, achieving comparable results with MST [47]. The other line of work exploited a plug-and-play framework [54] by integrating deep denoiser into the optimization algorithm.

2) *Single Image Super-Resolution:* The observation model of single image super-resolution can also be described mathematically as (2). In single image super-resolution, $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{y} \in \mathbb{R}^M$ denote the vectorized form of the desired high-resolution (HR) image and the observed low-resolution (LR) image respectively, and $\mathbf{A} \in \mathbb{R}^{M \times N}$ denotes the measurement matrix implemented by a blurring operator and a downsampling operator, where $N = HWL$ ($L = 3$ for color image), $M = \frac{N}{s^2}$, and s denotes the scaling factor. Model-based SR methods often treated SR as a regularized optimization problem and adopted the hand-crafted priors as regularizers, such as TV priors [24] and sparsity priors [18], [51], [55], [56], [57]. Inspired by the great success of SRCNN [58], learning-based SR methods have advanced rapidly and achieved amazing performance in recent years. Numerous learning-based SR methods designed elaborate networks to learn the non-linear mappings between LR images and HR images, e.g., Enhanced Deep Residual Networks (EDSR) [31], Residual Dense Network (RDN) [33], Residual Non-local Attention Networks (RNAN) [34], and Residual Channel Attention Network (RCAN) [36]. In parallel, the unfolding methods, e.g., Denoising Prior driven Deep Neural Network (DPDNN) [32] and Model-Guided Deep Unfolding Network (MoG-DUN) [35], have also shown comparable results and good generalization performance. SwinIR [46] proposed using Swin Transformer layers [49] to learn long-range dependencies of the images and achieved state-of-the-art SISR results. Both model-based SISR methods and learning-based SISR methods bear the same concerns with HSI reconstruction methods.

B. Vision Transformer for Image Restoration

Recently, the Vision Transformer (ViT) [59] proposed splitting the whole image into patches spatially and adopted self-attention layers to learn global similarities among patches. Due to the powerful capability for learning distant dependencies, numerous follow-up works based on ViT were proposed and achieved promising results in various vision tasks, e.g., image classification [49], [60], [61], [62], object detection [63], [64], [65], segmentation [66], [67], [68], etc. Motivated by the great success in high-level vision, Transformer-based methods for low-level vision have been proposed. In [69], a novel multi-head self-attention module and a novel feed-forward network, which are key modules in Transformer, have been designed to learn distant image dependencies. SwinIR [46] proposed

extracting image features by stacking Swin Transformer layers [49] with residual connections, achieving state-of-the-art SISR performance. Uformer [70] proposed an effective and efficient encoder-decoder network based on Transformer for noise, blur, and rain removal. To learn spectral-wise self-attention efficiently, MST [47] first attempted to introduce a Transformer into HSI reconstruction, leading to state-of-the-art HSI reconstruction performance.

III. DEEP GAUSSIAN SCALE MIXTURE PRIOR FOR IMAGE RECONSTRUCTION

This section extends the proposed method of the preliminary conference version [10] from SCI to image reconstruction. Details will be introduced, including GSM models for image reconstruction in Section III-A, deep GSM for image reconstruction in Section III-B, and network architecture of DGSM in Section III-C.

A. GSM Models for Image Reconstruction

In this paper, image reconstruction is formulated as a maximum a posteriori (MAP) estimation problem. In this manner, reconstructing the original image $\mathbf{x}$ from the observed measurement $\mathbf{y}$ involves maximizing the posterior

$$\log p(\mathbf{x}|\mathbf{y}) \propto \log p(\mathbf{y}|\mathbf{x}) + \log p(\mathbf{x}), \quad (3)$$

where $p(\mathbf{y}|\mathbf{x})$ and $p(\mathbf{x})$ denote the likelihood term and the (to be determined) prior distribution of $\mathbf{x}$, respectively. We adopt a Gaussian function to model the likelihood term $p(\mathbf{y}|\mathbf{x})$ as

$$p(\mathbf{y}|\mathbf{x}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2}{2\sigma^2}\right), \quad (4)$$

and model $\mathbf{x}$ by the GSM model that characterizes each pixel x_i with a *nonzero-mean* Gaussian distribution of standard deviation θ_i . Assuming that x_i and θ_i are independent, the prior term $p(\mathbf{x})$ can be expressed as

$$p(\mathbf{x}) = \prod_i p(x_i), \quad p(x_i) = \int_0^\infty p(x_i|\theta_i)p(\theta_i)d\theta_i, \quad (5)$$

where $p(\theta_i)$ denotes the scale prior, $p(x_i|\theta_i)$ is a nonzero-mean Gaussian distribution with variance θ_i^2 and mean u_i , i.e.,

$$p(x_i|\theta_i) = \frac{1}{\sqrt{2\pi}\theta_i} \exp\left(-\frac{(x_i - u_i)^2}{2\theta_i^2}\right). \quad (6)$$

Adopting different scale priors, we can express the GSM model as different distributions.

For the scale prior $p(\theta_i)$, existing GSM methods [18], [71], [72] adopted the Jeffrey's prior (i.e., $p(\theta_i) = \frac{1}{\theta_i}$). Unlike existing GSM methods with an exact prior, we introduce a general form with an energy function $J(\theta_i)$, as

$$p(\theta_i) \propto \exp(-J(\theta_i)). \quad (7)$$

Although both $p(x_i|\theta_i)$ and $p(\theta_i)$ are given, the analytical expression of $p(x_i)$ (i.e., (5)) is hard to compute. Therefore, we propose jointly estimating $\mathbf{x}$ and $\boldsymbol{\theta}$ by replacing $p(\mathbf{x})$ with

$p(\mathbf{x}, \boldsymbol{\theta})$ in the MAP estimator as follows,

$$\begin{aligned} (\mathbf{x}, \boldsymbol{\theta}) &= \arg \max_{\mathbf{x}, \boldsymbol{\theta}} \log p(\mathbf{y}|\mathbf{x}) + \log p(\mathbf{x}, \boldsymbol{\theta}) \\ &= \arg \max_{\mathbf{x}, \boldsymbol{\theta}} \log p(\mathbf{y}|\mathbf{x}) + \log p(\mathbf{x}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta}). \end{aligned} \quad (8)$$

By substituting the Gaussian likelihood term of (4) and the prior terms of $p(x_i|\theta_i)$ and $p(\theta_i)$ into the above MAP estimator, we obtain the minimization

$$\begin{aligned} (\mathbf{x}, \boldsymbol{\theta}) &= \arg \min_{\mathbf{x}, \boldsymbol{\theta}} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \sigma^2 \sum_{i=1}^N \frac{1}{\theta_i^2} (x_i - u_i)^2 \\ &\quad + 2\sigma^2 \sum_{i=1}^N \log \theta_i + 2\sigma^2 J(\boldsymbol{\theta}) \\ &= \arg \min_{\mathbf{x}, \boldsymbol{\theta}} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \sigma^2 \sum_{i=1}^N \frac{1}{\theta_i^2} (x_i - u_i)^2 \\ &\quad + R(\boldsymbol{\theta}), \end{aligned} \quad (9)$$

where $R(\boldsymbol{\theta}) = 2\sigma^2 \sum_{i=1}^N \log \theta_i + 2\sigma^2 J(\boldsymbol{\theta})$. The objective function can be split into a $\mathbf{x}$ -subproblem and a $\boldsymbol{\theta}$ -subproblem and can be solved by alternating optimizing these two subproblems.

With fixed $\boldsymbol{\theta}$, the $\mathbf{x}$ -subproblem can be written as

$$\mathbf{x} = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \sum_{i=1}^N w_i (x_i - u_i)^2, \quad (10)$$

where $w_i = \frac{\sigma^2}{\theta_i^2}$ and the mean u_i keeps updating with $\mathbf{x}$. The first term and the second term of (10) are the data fidelity term and the regularization term, respectively. Therefore, the spatially-variant $\mathbf{w}$ can be viewed as a regularization parameter for balancing the data fidelity term and the regularization term. Inspired by the auto-regressive (AR) model [17], we can predict the mean u_i by calculating the weighted average of the local spatial-spectral neighboring pixels. Some existing methods (e.g., the guided filtering [73], [74], the nonlocal means methods [75], [76] or the deep learning based method [77]) can be used to estimate the local spatial-spectral similarity weights. Therefore, (10) can be solved by a single-step gradient descent, as

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - 2\delta \{ \mathbf{A}^\top (\mathbf{A}\mathbf{x}^{(t)} - \mathbf{y}) + \mathbf{w}^{(t)} (\mathbf{x}^{(t)} - \mathbf{u}^{(t)}) \}, \quad (11)$$

where $\mathbf{u}^{(t)} = [u_1^t, \dots, u_N^t]^\top \in \mathbb{R}^N$, $\mathbf{w}^{(t)} = [w_1^t, \dots, w_N^t]^\top \in \mathbb{R}^N$ and δ is the step size.

The $\boldsymbol{\theta}$ -subproblem can be changed to estimate $\mathbf{w}$. With fixed $\mathbf{x}$, $\mathbf{w}$ can be estimated by

$$\mathbf{w} = \arg \min_{\mathbf{w}} \sum_{i=1}^N w_i (x_i - u_i)^2 + R(\mathbf{w}). \quad (12)$$

With different $R(\mathbf{w})$, $\mathbf{w}$ can be estimated by different ways. For some $R(\mathbf{w})$, a closed-form solution can be achieved [71]; for others, iterative algorithms might be used. However, each of them has their pros and cons. To cope with this challenge, instead of using a manually designed proximal operator, we estimate

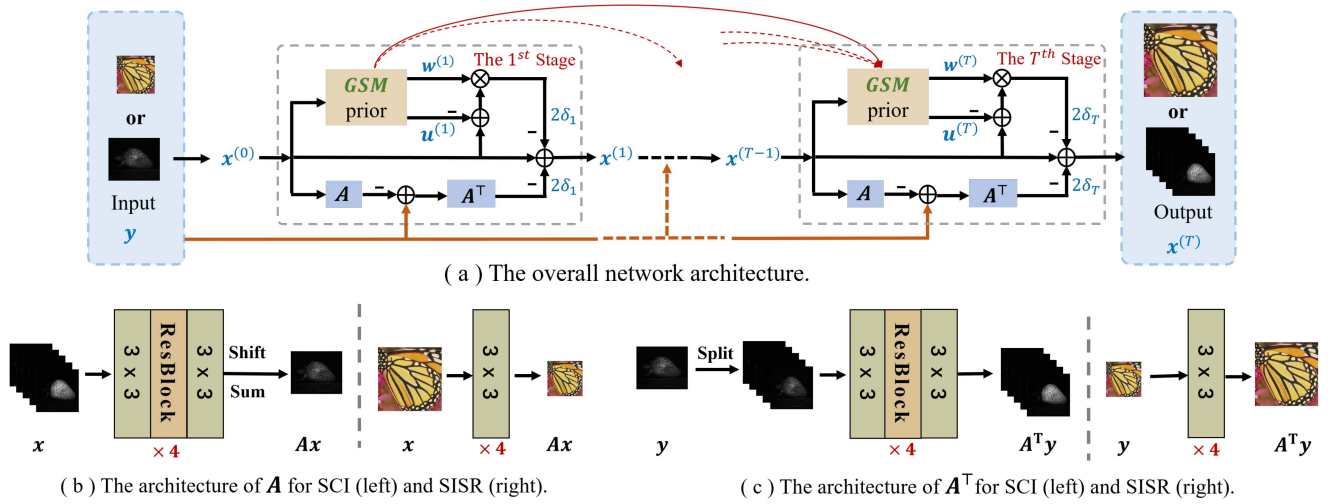


Fig. 2. Architecture of the proposed network DGSM for spectral compressive imaging (SCI) and single image super-resolution (SISR). The architectures of (a) the overall network, (b) the measurement matrix for SCI (left) and SISR (right), (c) the transposed version of the measurement matrix for SCI (left) and SISR (right).

$w^{(t+1)}$ from $x^{(t+1)}$ using a deep network, as described in the next subsection.

B. Deep GSM for Image Reconstruction

In general, alternating optimizing x and w requires numerous iterations to converge and is time-consuming. Moreover, it also requires a hand-crafted prior $p(\theta)$ and all the algorithm parameters cannot be jointly optimized, leading to limited performance. To address these issues and improve reconstruction quality, we optimize x and w jointly via a deep network. For network design purpose, we recombine the x and w -subproblems via a united framework

$$x^{(t+1)} = x^{(t)} - 2\delta \left\{ A^T \left(Ax^{(t)} - y \right) + S \left(x^{(t)} \right) \left(x^{(t)} - u^{(t)} \right) \right\}, \quad (13)$$

where $S(\cdot)$ represents the function of the deep network for estimating w , i.e., the solution of (12). As shown in Fig. 2(a), we construct the end-to-end network with T stages corresponding to T iterations for iteratively optimizing x and w . The proposed network, named DGSM, is applied to two subtasks (i.e., SCI and SISR) and consists of the following main modules.

- In SCI, the measurement y is split into a 3D data cube of size $H \times W \times L$ to initialize x . In SISR, we use a bilinear interpolation operator to initialize x .
- We use two sub-networks to learn the measurement matrix A and its transposed version A^T .
- In the previous version [10], we proposed using a U-Net to estimate u and w . In the present version, for learning long-range dependencies of pixels, we develop an enhanced variant based on Swin Transformer to estimate u and w more accurately. The methods proposed in the previous version and the present version are named as DGSM-UNet and DGSM-Swin, respectively.

C. Network Architecture

1) *Modules for Learning the Measurement Matrices:* Considering that real-world images have a large spatial size and the measurement matrices are often sophisticated, network training with explicitly constructed A and A^T requires a large amount of GPU memory and computational complexity. To address this issue, we propose two sub-networks for learning these two operations, respectively. The measurement matrix A represents a hybrid operator, e.g., shifting and summation in SCI and blurring and down-sampling in SISR.

For spectral compressive imaging (SCI) [78] reconstruction, the proposed method of the previous version [10] adopts two Conv layers and four ResBlocks to implement the measurement matrix A and A^T , respectively. Each ResBlock [79] consists of two Conv layers with a ReLU nonlinearity function plus a skip connection. Regarding the measurement matrix A , as shown in Fig. 2(b), x is fed into the sub-network to generate modulated feature maps that are further shifted and summed along the spectral dimension to generate the measurements $y = Ax$. Regarding A^T , as shown in Fig. 2(c), we first slide an $H \times W$ extraction window on the input y of size $H \times (W + L - 1)$ with the slide step one pixel and split the input into an L -channel image of size $H \times W$. Then the split sub-images are fed into the sub-network to generate the estimate $A^T y$.

For single image super-resolution (SISR), the measurement matrix A can be implemented by three Conv layers with stride 1 and a Conv layer with the stride s to downsample the spatial resolution as shown in Fig. 2(b). Inversely, A^T can also be implemented by three Conv layers with stride 1 and a Deconv layer with stride s to upsample the spatial resolution as shown in Fig. 2(c).

2) *Transformer-Based GSM Prior Network:* In the previous version [10], we adopted a U-Net based on 3×3 Conv layers to learn the GSM prior, i.e., the variances and the means, which neglects the long-range dependencies of images resulting in

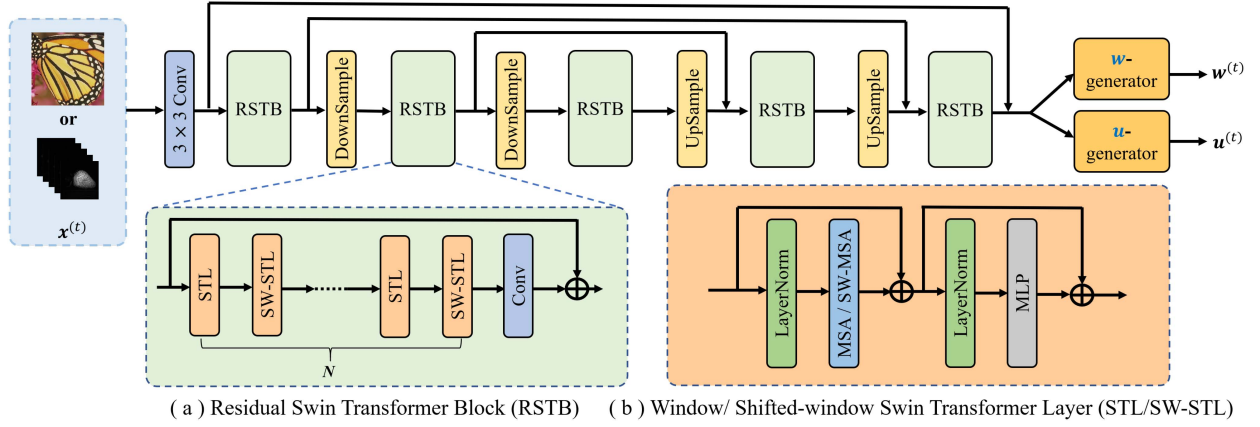


Fig. 3. Architecture of the proposed Transformer-based network for learning the GSM priors. (a) Residual Swin Transformer Block (RSTB). (b) Window/Shifted-window Swin Transformer Layer (STL/SW-STL).

limited reconstruction quality. Numerous image reconstruction methods [35], [39] have verified that capturing long-range dependencies is helpful to improve reconstruction quality. Recently, Transformer-based methods [59] have shown powerful long-range dependency learning capability by the self-attention mechanism. Furthermore, Swin Transformer [49] employed the self-attention mechanism in local windows to reduce the computation and the shift windows operator to enlarge the receptive field, leading to long-range dependency learning in a progressive manner. To improve the representation learning of DGSM, we employ a Transformer network based on Swin Transformer instead of the U-Net to estimate the variances and the means of the GSM models.

This subsection describes the adopted Transformer-based network to learn GSM prior in detail as shown in Fig. 3. The Transformer-based GSM prior network consists of an embedding layer, two encoding blocks, a bottleneck block, two decoding blocks, a w -generator, and an u -generator. The embedding layer is a 3×3 Conv layer and outputs the embedded features with 64 channels. Both the encoding and decoding block contain a Residual Swin Transformer Block (RSTB) which consists of P (e.g., 2, 4, or 6) Swin Transformer layers, a 3×3 Conv layer, and a residual connection as shown in Fig. 3(a). Similar to the U-Net, every encoding block is followed by a downsampling layer with a stride of 2 to downsample the feature map, and inserts an upsampling layer with a stride of 2 ahead of every decoding block to increase the spatial resolutions of the feature maps. The outputs of the encoding block will be fused at the decoding block. Given the input $h \times w \times c$, the Swin Transformer layer will divide the input into $\frac{hw}{m^2}$ local non-overlapping windows of size $m \times m \times c$. Then, the multi-head self-attention (MSA) mechanism will be employed to learn the dependencies in each local window. As shown in Fig. 3(b), the output of the Swin Transformer layer can be computed as

$$\begin{aligned} \mathbf{X}' &= \text{MSA}(\text{LN}(\mathbf{X})) + \mathbf{X}, \\ \mathbf{X}'' &= \text{FFN}(\text{LN}(\mathbf{X}')) + \mathbf{X}', \end{aligned} \quad (14)$$

where $\mathbf{X}, \mathbf{X}', \mathbf{X}'' \in \mathbb{R}^{m^2 \times c}$ denotes the local windows, $\text{MSA}(\cdot)$ and $\text{LN}(\cdot)$ denote the multi-head self-attention mechanism and

the LayerNorm layer, and $\text{FFN}(\cdot)$ denotes a multi-layer perceptron (MLP) consisting of two fully-connected layers and a GELU non-linearity layer. In the multi-head self-attention mechanism, three learnable matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{c \times c}$ are used to project the local window $\mathbf{X}$ into *query* matrix $\mathbf{Q} \in \mathbb{R}^{m^2 \times c}$, *key* matrix $\mathbf{K} \in \mathbb{R}^{m^2 \times c}$, and *value* matrix $\mathbf{V} \in \mathbb{R}^{m^2 \times c}$ as

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \mathbf{K} = \mathbf{X}\mathbf{W}_K, \mathbf{V} = \mathbf{X}\mathbf{W}_V. \quad (15)$$

The whole process of the multi-head self-attention mechanism with n heads can be represented as

$$\text{MSA}(\mathbf{X}) = \text{Concat}_{j=1}^n \left(\text{SoftMax}(\mathbf{Q}_j \mathbf{K}_j^\top / \sqrt{\frac{c}{n}} + \mathbf{B}_j) \mathbf{V}_j \right), \quad (16)$$

where $\mathbf{Q}_j, \mathbf{K}_j, \mathbf{V}_j \in m^2 \times \frac{c}{n}$ denote the j^{th} head of $\mathbf{Q}, \mathbf{K}, \mathbf{V}$, $\text{SoftMax}(\cdot)$ denotes the softmax operator, $\mathbf{B}_j$ denotes the relative positional encoding, and $\text{Concat}(\cdot)$ concatenates the outputs of all heads. To learn long-range dependencies, the shifting window operator has been employed in the multi-head self-attention mechanism. Both the w -generator and the u -generator are a single 3×3 Conv layer to estimate the regularization parameters w and the means u , respectively. We have also shown several learned regularization parameter maps w by the proposed Transformer-based GSM prior network in Figs. 4 (bottom) and 5 (right). From Figs. 4 (bottom) and 5 (right), we can see that the well-learned w are also consistent with the image edges and textures. Although the w learned by DGSM-UNet varies spatially, it loses the ability to distinguish in the spectral dimension. In contrast, the w learned by DGSM-Swin vary not only spatially but also spectrally and matches with the content of images, verifying the efficiency of DGSM-Swin.

IV. SIMULATION RESULTS

A. Experimental Setup

To verify the performance of the proposed image reconstruction method, we conduct extensive experiments on two subtasks (i.e., SCI and SISR). The peak-signal-to-noise (PSNR) and the

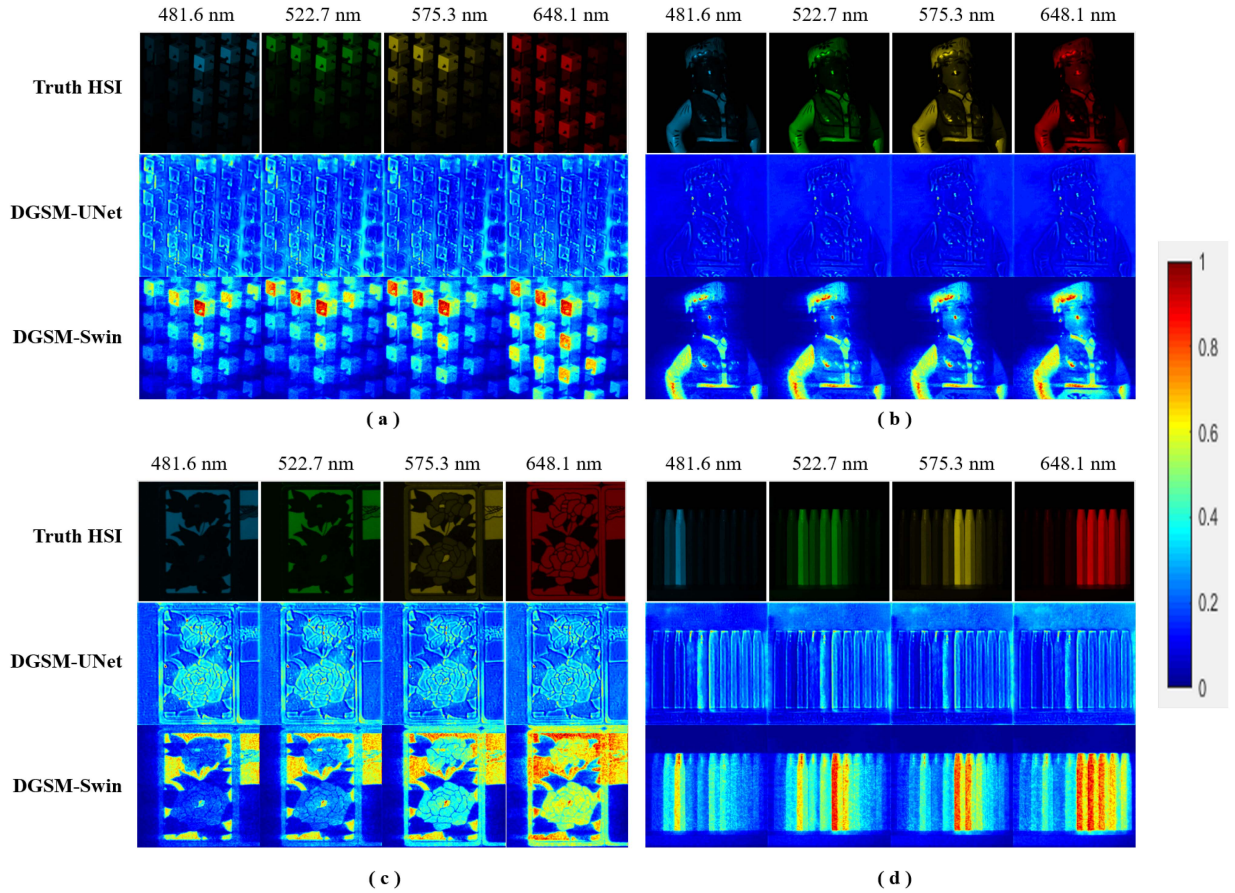


Fig. 4. The visualization (with normalization) of the regularization parameters w learned by *DGSM-UNet* [10] (middle) and *DGSM-Swin* (bottom) for SCI. (a) Scene 2. (b) Scene 6. (c) Scene 7. (d) Scene 9.

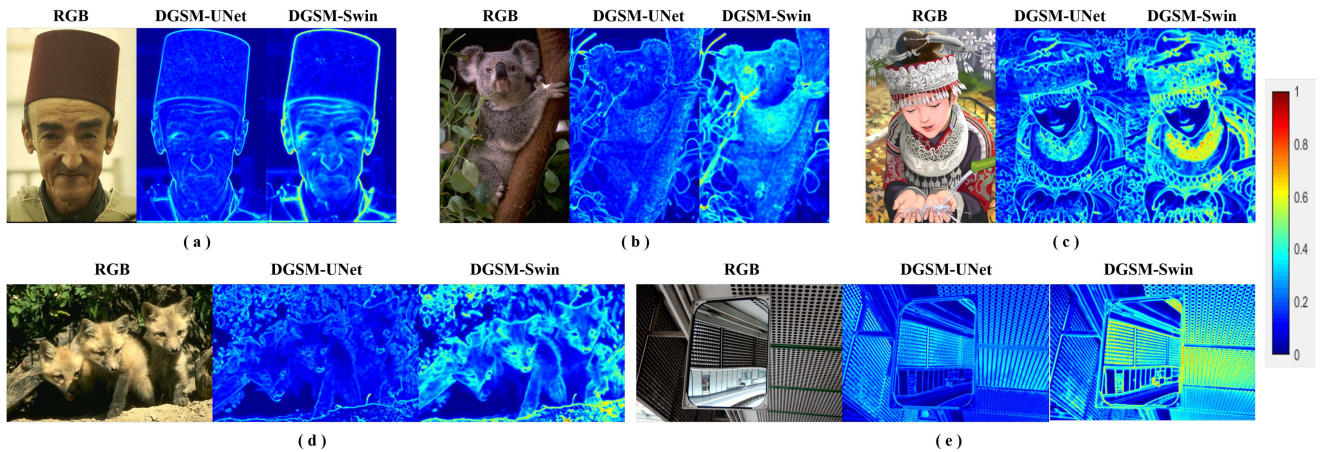


Fig. 5. The visualization (with normalization) of the regularization parameters w learned by *DGSM-UNet* (middle) and *DGSM-Swin* (right) for SISR. (a) *img_035* from BSD100. (b) *img_088* from BSD100. (c) *img_005* from Set14. (d) *img_025* from BSD100. (e) *img_004* from Urban100.

structural similarity index (SSIM) [80] are both employed to evaluate the performance of the image reconstruction methods.

1) *Dataset*: For SCI, extensive simulation experiments on two public HSI datasets, CAVE [81] and KAIST [82], have been conducted. The CAVE dataset consists of 32 HSIs of spatial size 512×512 with 31 spectral bands. The KAIST

dataset has 30 HSIs of spatial size 2704×3376 also with 31 spectral bands. Similar to TSA-Net [20], we employ the *real mask* of size 256×256 for simulation. Following the procedure in TSA-Net [20], we train the proposed network on the CAVE dataset, and test on 10 scenes of spatial size 256×256 extracted from the KAIST dataset. To be consistent with the

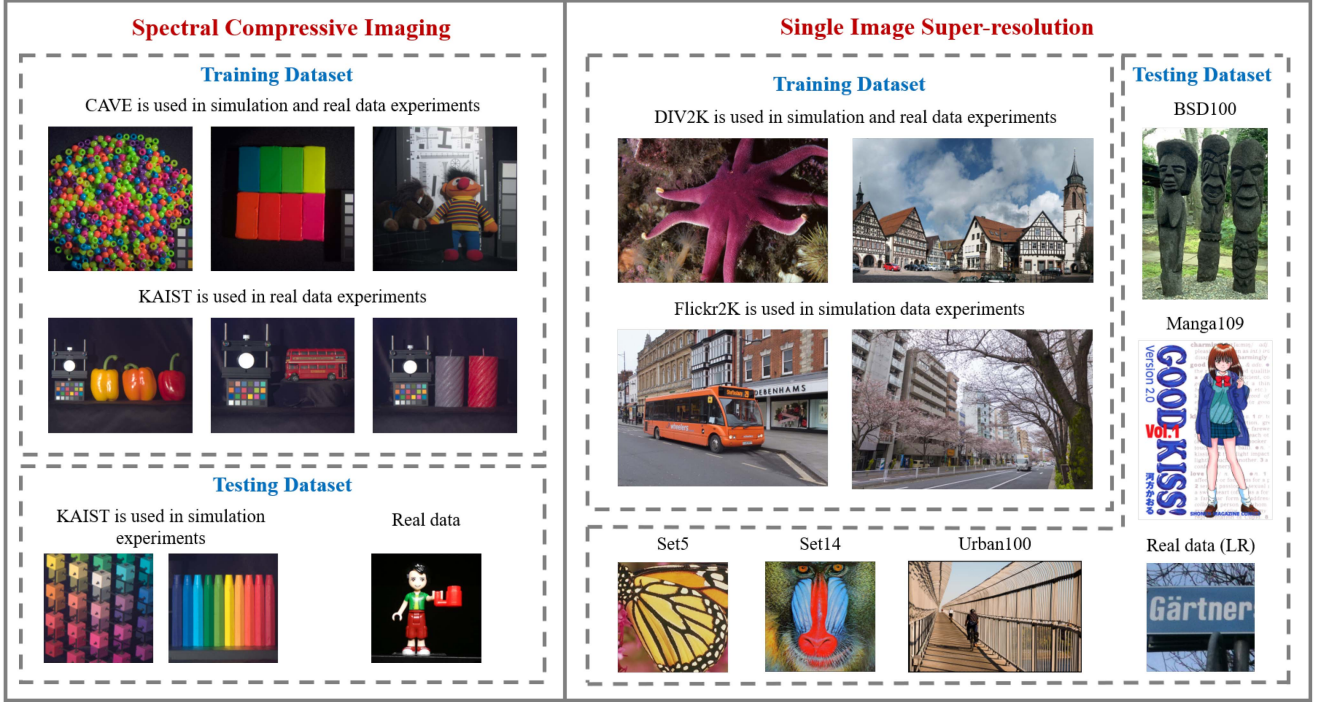


Fig. 6. The overview of the used SCI and SISR datasets in simulation and real data experiments. In spectral compressive imaging, we show the corresponding RGB images of HSIs.

wavelength of the real system [20], we unify the wavelength of the training and testing data by spectral interpolation. Thus, the modified training and testing data have 28 spectral bands ranging from 450 nm to 650 nm. During training, to simulate the measurements following (1), we first randomly extract $H \times W \times 28$ (e.g., $H = W = 96$ or 256) patches from the training dataset as training labels (ground truth HSI) and randomly extract $H \times W$ patches from the *real mask* as training masks to generate the modulated data of size $H \times W \times 28$. Then the modulated data is shifted in spatial dimension at an interval of two pixels for generating the dispersed data of size $H \times (W + 54) \times 28$. At last, the spectral dimension of the dispersed data is summed up to generate the 2D measurements of size $H \times (W + 54)$ as the network inputs. We use random flipping and rotation for data augmentation.

For SISR, following [35], [36], we train the proposed networks (i.e., DGSM-UNet and DGSM-Swin) on DIV2K [83] and Flickr2K [84], and test on five standard benchmark datasets -i.e., Set5 [85], Set14 [86], BSD100 [87], Urban100 [88], and Manga109 [89]. During training, we first randomly extract $H \times W \times 3$ (e.g., $H = W = 96, 128$, or 192) high-resolution (HR) patches from the training dataset as training labels and then apply bicubic downsampling to the HR images to obtain the low-resolution (LR) inputs. The scale factors are set to 2, 3, and 4, respectively.

In Fig. 6, we show the overview of the used SCI and SISR datasets in simulation experiments. For SCI, we show the corresponding RGB images of HSIs.

2) *Implementation Details*: We implement the proposed method in PyTorch and set $T = 4$ in the DGSM-UNet [10]

implementation and $T = 2$ and $P = 4$ in the DGSM-Swin implementation for achieving a good trade-off between reconstruction performance and computational complexity. We train the proposed DGSM method in an end-to-end manner using four Nvidia Titan XP GPUs. The learning rate is set to 10^{-4} . The step size δ in different stages is learnable and is initialized as 0.1 empirically. Except for δ , all the network parameters of each stage are shared. During training with D training samples, the network parameters Θ can be optimized by minimizing the loss function

$$\hat{\Theta} = \arg \min_{\Theta} \frac{1}{D} \sum_{d=1}^D \|\mathcal{F}(\mathbf{y}_d; \Theta) - \mathbf{x}_d\|_1, \quad (17)$$

where $\mathbf{y}_d$ and $\mathbf{x}_d$ denote the d^{th} pair of observation and ground-truth respectively, and $\mathcal{F}(\mathbf{y}_d; \Theta)$ represents the output of the proposed network. Since the ℓ_1 norm has been proven to be better than the ℓ_2 norm in preserving image edges and textures [1], [90], [91], we adopt the ℓ_1 norm in the loss function. The ADAM optimizer [92] with setting $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$ is exploited to train the proposed network DGSM.

B. Comparison With State-of-the-Art Spectral Compressive Imaging Methods

We employ the proposed image reconstruction method DGSM to the SCI task and compare it with several state-of-the-art SCI methods, including three model-based methods (i.e., TwIST [51], GAP-TV [25] and DeSCI [9]) and seven deep learning-based methods (i.e., λ -net [37], DSSP [38], DNU [39], TSA-Net [20], HDNet [44], GAP-CCoT [23], and MST [47]). As the source code is unavailable, we re-implemented DSSP and

TABLE I

THE PSNR IN DB (TOP ENTRY IN EACH CELL) AND SSIM (BOTTOM ENTRY IN EACH CELL) RESULTS OF THE TEST METHODS ON 10 SCENES. THE BEST PERFORMANCE IS SHOWN IN **BOLD** AND THE SECOND BEST PERFORMANCE IS SHOWN IN UNDERLINE

Method	TwIST [51]	GAP-TV [25]	DeSCI [9]	λ -net [37]	DSSP [38]	DNU [39]	TSA-Net [20]	HDNet [44]	GAP-CCoT [23]	MST [47]	DGSM-UNet [10]	DGSM-Swin
Scene1	25.16 0.700	26.82 0.754	27.13 0.748	30.10 0.849	31.48 0.858	31.72 0.863	32.03 0.892	34.95 0.948	35.13 0.936	35.40 0.941	33.26 0.915	36.40 0.952
Scene2	23.02 0.604	22.89 0.610	23.04 0.620	28.49 0.805	31.09 0.842	31.13 0.846	31.00 0.858	32.52 <u>0.953</u>	35.83 0.945	35.87 0.944	32.09 0.898	37.14 0.958
Scene3	21.40 0.711	26.31 0.802	26.62 0.818	27.73 0.870	28.96 0.823	29.99 0.845	32.25 0.915	34.52 <u>0.957</u>	36.78 0.956	36.51 0.953	33.06 0.925	38.16 0.963
Scene4	30.19 0.851	30.65 0.852	34.96 0.897	37.01 0.934	34.56 0.902	35.34 0.908	39.19 0.953	43.00 <u>0.981</u>	41.73 0.971	42.27 0.973	40.54 0.964	44.41 0.982
Scene5	21.41 0.635	23.64 0.703	23.94 0.706	26.19 0.817	28.53 0.808	29.03 0.833	29.39 0.884	32.49 <u>0.957</u>	32.57 0.945	32.77 0.947	28.86 0.882	33.89 0.958
Scene6	20.95 0.644	21.85 0.663	22.38 0.683	28.64 0.853	30.83 0.877	30.87 0.887	31.44 0.908	35.96 0.965	34.87 0.954	34.80 0.955	33.08 0.937	35.79 <u>0.964</u>
Scene7	22.20 0.643	23.76 0.688	24.45 0.743	26.47 0.806	28.71 0.824	28.99 0.839	30.32 0.878	29.18 <u>0.937</u>	33.43 0.924	33.66 0.925	30.74 0.886	35.16 0.945
Scene8	21.82 0.650	21.98 0.655	22.03 0.673	26.09 0.831	30.09 0.881	30.13 0.885	29.35 0.888	34.00 <u>0.961</u>	33.11 0.948	32.67 0.948	31.55 0.923	34.20 0.962
Scene9	22.42 0.690	22.63 0.682	24.56 0.732	27.50 0.826	30.43 0.868	31.03 0.876	30.01 0.890	34.56 <u>0.958</u>	35.69 0.951	35.39 0.949	31.66 0.911	37.21 0.959
Scene10	22.67 0.569	23.10 0.584	23.59 0.587	27.13 0.816	28.78 0.842	29.14 0.849	29.59 0.874	32.22 0.950	32.41 0.937	32.50 0.941	31.44 0.925	33.17 <u>0.947</u>
Average	23.13 0.669	24.36 0.699	25.27 0.721	28.53 0.841	30.35 0.852	30.74 0.863	31.46 0.894	34.34 <u>0.957</u>	35.16 0.947	35.18 0.948	32.63 0.917	36.55 0.959

DNU by ourselves. For other competing methods, the source code or reconstruction results used in this comparison were released by their authors. For the sake of fair comparison, all deep learning methods were *trained on the same training dataset*. The reconstruction results of all competing methods on the 10 testing scenes are shown in Table I. We can observe that the deep learning-based methods outperform the model-based methods by a large margin, and the proposed method DGSM-Swin significantly outperforms other deep learning-based methods. Specifically, compared with the second-best method MST [47], the third-best method GAP-CCoT [23], and the fourth-best method HDNet [44], the improvements by the proposed network DGSM-Swin over MST, GAP-CCoT and HDNet are 1.37 dB, 1.39 dB, and 2.21 dB in average, respectively. Our proposed networks DGSM-UNet and DGSM-Swin outperform the method TSA-Net [20] by 1.17 dB and 5.10 dB in average PSNR and 0.023 and 0.065 in average SSIM, respectively. Compared with the two deep unfolding methods DSSP [38] and DNU [39], the improvements by the proposed methods (DGSM-UNet and DGSM-Swin) over DSSP and DNU are (2.28 dB and 6.20 dB) and (1.89 dB and 5.81 dB) in average, respectively. The DSSP and DNU methods only estimated the image mean (i.e., the denoising prior) but neglected the variance, resulting in severe blurring artifacts. In contrast, the proposed methods DGSM-UNet and DGSM-Swin have advantages in adapting to various HSI edges and textures and can reconstruct more edge details due to the well-learned variances. The proposed method DGSM-Swin outperforms DGSM-UNet by 3.92 dB in average PSNR and 0.042 in average SSIM, verifying the efficiency of DGSM-Swin. Fig. 7 plots selected frames and spectral curves of the reconstructed HSIs by the eight deep learning-based methods. We can see that the HSIs reconstructed by the proposed methods DGSM-UNet and DGSM-Swin have sharper edges and textures and less undesirable visual artifacts than the other methods.

C. Comparison With State-of-the-Art Single Image Super-Resolution Methods

We also employ the proposed methods DGSM-UNet and DGSM-Swin to the SISR task and compare them with several

state-of-the-art SISR methods, including seven deep learning-based methods (i.e., EDSR [31], DPDNN [32], RDN [33], RNAN [34], MoG-DUN [35], RCAN [36], and SwinIR [46]). Table II shows the average PSNR and SSIM results for bicubic downsampling degradation on five benchmark datasets by nine competing methods. From Table II, it is easy to see that our method DGSM-Swin is superior to DGSM-UNet in terms of the average PSNR and SSIM results. Compared with DGSM-UNet, the improvements on the Urban100 testset by DGSM-Swin over DGSM-UNet are 0.65 dB, 1.47 dB, and 0.20 dB with the scaling factors of $\times 2$, $\times 3$, and $\times 4$, respectively. When compared with most competing methods, our method DGSM-Swin achieves highly comparable and sometimes even better results. Although SwinIR [46] achieves the best results, the number of FLOPS of SwinIR (67.50 G FLOPS) is nearly twice that of DGSM-Swin (35.25 G FLOPS) evaluated on the LR input of size $64 \times 64 \times 3$ and the scaling factor of $\times 2$. In addition, our method DGSM-Swin and SwinIR achieve better results by enlarging the training dataset (i.e., DIV2K+Flickr2K). Figs. 8 and 9 show the visual quality comparison results of the competing methods with the scaling factors of $\times 3$, and $\times 4$, respectively. We can see that the proposed methods DGSM-UNet and DGSM-Swin can reconstruct more details around edges and textures and better suppress undesirable visual artifacts, while most competing methods suffer from severe blurring or over smooth artifacts.

D. Ablation Study

We have conducted several ablation studies on SCI to verify the impacts of the w -generator and the number of stages T and Swin Transformer Layers (STLs) P .

Table III shows the comparison results of the proposed networks DGSM-UNet and DGSM-Swin with and without the w -generator. From Table III, we can observe that using the w -generator in DGSM-UNet and DGSM-Swin can improve the reconstruction results by 0.80 dB and 0.62 dB, respectively. As shown in Fig. 4, the well-learned w generated by the w -generator will guide the proposed network to pay attention to the image edges and textures, leading to better reconstruction performance.

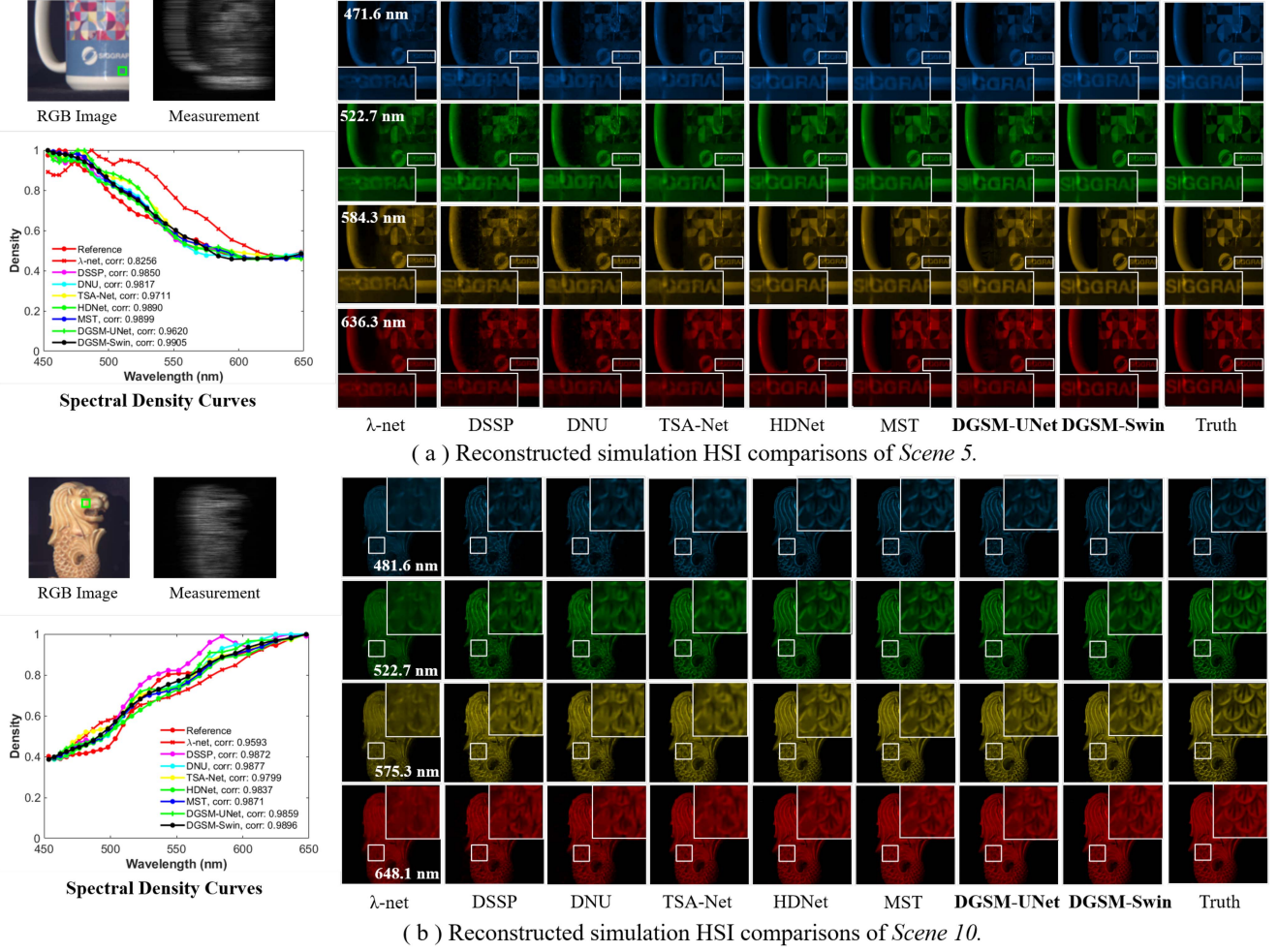


Fig. 7. Reconstructed images of *Scene 5* (a) and *Scene 10* (b) with 4 out of 28 spectral channels by the eight deep learning-based methods. The spectral curves (bottom-left) correspond to the selected green box of the RGB image. Zoom in for better view.

To explore the impact of the number of stages T and Swin Transformer Layers (STLs) P for SCI, we have also conducted ablation studies on the proposed DGSM-Swin. Table IV shows the reconstruction results under different network settings. From Table IV, we can observe that increasing the number of stages T and STLs P leads to better results. The improvement flattens out after $T = 2$ and $P = 4$. Therefore, to achieve a good trade-off between reconstruction performance and computational complexity, we set $T = 2$ and $P = 4$ in our implementation. Compared with using four stages in DGSM-UNet, using one or two stages in DGSM-Swin is sufficient and achieves state-of-the-art results.

In addition, considering that (10) can be solved via an exact solution as

$$\mathbf{x}^{(t+1)} = (\mathbf{A}^\top \mathbf{A} - \mathbf{W})^{-1} (\mathbf{A}^\top \mathbf{y} + \mathbf{W} \mathbf{u}^t), \quad (18)$$

where $\mathbf{x} \in \mathbb{R}^N$, $\mathbf{u} \in \mathbb{R}^N$, $\mathbf{y} \in \mathbb{R}^M$, and $\mathbf{W} = \text{diag}\{w_1, w_2, \dots, w_N\}$. In SCI, due to the specific structure of the measurement matrix $\mathbf{A}$, we can simplify the exact solution. Similar to [9], [25], [39], we can obtain the simplified solution

as

$$\mathbf{x}^{(t+1)} = \mathbf{u}^{(t)} + \mathbf{A}^\top \left[(\mathbf{y} - \mathbf{A} \mathbf{u}^{(t)}) \oslash (\mathbf{W}' + \mathbf{A} \mathbf{A}^\top) \right], \quad (19)$$

where $\mathbf{W}' = \text{diag}\{w_1, w_2, \dots, w_M\}$ and $\oslash$ denotes the element-wise division. In Table IV, we compare the results of using gradient descent and using the simplified exact solution. From Table IV, we can observe that using gradient descent can achieve better results than using the exact solution. With 2 stages and 4 STLs, the improvements by using gradient descent over using the exact solution are 0.33 dB in average PSNR and 0.011 in average SSIM. We analyze that the main reason is the manifold projection did not consider the noise during the iteration as pointed out in [40]. On the other hand, in SISR, the measurement matrices $\mathbf{A}$ and $\mathbf{A}^\top$ are often sophisticated, making $\mathbf{A}$ and $\mathbf{A}^\top$ hard to construct and $\mathbf{A}^\top \mathbf{A}$ hard to calculate. Therefore, the exact solution can't be obtained in SISR. In general, using gradient descent to solve (10) is suitable for most image reconstruction tasks. The exact solution could work in SCI, but not in SISR.

TABLE II

AVERAGE PSNR AND SSIM RESULTS FOR BICUBIC DOWNSAMPLING DEGRADATION ON FIVE BENCHMARK DATASETS. THE BEST PERFORMANCE IS SHOWN IN **BOLD** AND THE SECOND BEST PERFORMANCE IS SHOWN IN UNDERLINE

Method	Scale	Training Dataset	Set5 [85]		Set14 [86]		BSD100 [87]		Urban100 [88]		Manga109 [89]	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Bicubic	$\times 2$	-	33.66	0.9299	30.24	0.8488	29.56	0.8431	26.88	0.8403	30.80	0.9339
EDSR [31]	$\times 2$	DIV2K	38.11	0.9602	33.92	0.9195	32.32	0.9013	32.93	0.9351	39.10	0.9773
DPDNN [32]	$\times 2$	DIV2K	37.75	0.9600	33.30	0.9150	32.09	0.8990	31.50	0.9220	-	-
RDN [33]	$\times 2$	DIV2K	38.24	0.9614	34.01	0.9212	32.34	0.9017	32.89	0.9353	39.18	0.9780
RNAN [34]	$\times 2$	DIV2K	38.17	0.9611	33.87	0.9207	32.32	0.9014	32.73	0.9340	39.23	0.9785
MoG-DUN [35]	$\times 2$	DIV2K	38.25	0.9614	34.00	0.9205	32.37	0.9020	32.75	0.9421	39.37	0.9783
RCAN [36]	$\times 2$	DIV2K	<u>38.27</u>	0.9614	<u>34.12</u>	0.9216	<u>32.41</u>	<u>0.9027</u>	<u>33.34</u>	0.9384	<u>39.44</u>	<u>0.9786</u>
SwinIR [46]	$\times 2$	DIV2K	38.35	0.9620	34.14	0.9227	32.44	0.9030	33.40	0.9393	39.60	0.9792
DGSM-UNet [10]	$\times 2$	DIV2K	38.08	0.9610	33.80	0.9204	32.28	0.9015	32.30	0.9389	38.93	0.9777
DGSM-Swin	$\times 2$	DIV2K	38.24	0.9615	33.93	0.9217	32.36	0.9019	32.95	0.9442	39.31	0.9783
DBPN [93]	$\times 2$	DIV2K+Flickr2K	38.09	0.9600	33.85	0.9190	32.27	0.9000	32.55	0.9324	38.89	0.9775
SwinIR [46]	$\times 2$	DIV2K+Flickr2K	38.42	0.9623	34.46	0.9250	32.53	0.9041	33.81	0.9427	39.92	0.9797
DGSM-Swin	$\times 2$	DIV2K+Flickr2K	<u>38.30</u>	<u>0.9618</u>	<u>34.19</u>	<u>0.9230</u>	<u>32.43</u>	<u>0.9029</u>	<u>33.18</u>	0.9462	39.60	0.9790
Bicubic	$\times 3$	-	30.39	0.8682	27.55	0.7742	27.21	0.7385	24.46	0.7349	26.95	0.8556
EDSR [31]	$\times 3$	DIV2K	34.65	0.9280	30.52	0.8462	29.25	0.8093	28.80	0.8653	34.17	0.9476
DPDNN [32]	$\times 3$	DIV2K	33.93	0.9240	30.02	0.8360	29.00	0.8010	27.61	0.8420	-	-
RDN [33]	$\times 3$	DIV2K	34.71	0.9296	30.57	0.8468	29.26	0.8093	28.80	0.8653	34.13	0.9484
RNAN [34]	$\times 3$	DIV2K	34.66	0.9284	30.52	0.8466	29.26	0.8099	28.75	0.8649	34.25	0.9486
MoG-DUN [35]	$\times 3$	DIV2K	34.76	0.9300	30.63	0.8479	29.24	0.8094	28.82	0.8651	34.34	0.9490
RCAN [36]	$\times 3$	DIV2K	34.74	0.9299	<u>30.65</u>	0.8482	<u>29.32</u>	<u>0.8111</u>	<u>29.09</u>	<u>0.8702</u>	<u>34.44</u>	<u>0.9499</u>
SwinIR [46]	$\times 3$	DIV2K	34.89	0.9312	30.77	0.8503	29.37	0.8124	29.29	0.8744	34.74	0.9518
DGSM-UNet [10]	$\times 3$	DIV2K	34.28	0.9239	30.41	0.8428	28.97	0.8027	27.46	0.8383	32.89	0.9396
DGSM-Swin	$\times 3$	DIV2K	34.77	0.9300	30.65	0.8490	29.29	0.8109	28.93	0.8684	34.30	0.9498
SwinIR [46]	$\times 3$	DIV2K+Flickr2K	34.97	0.9318	30.93	0.8534	29.46	0.8145	29.75	0.8826	35.12	0.9537
DGSM-Swin	$\times 3$	DIV2K+Flickr2K	34.83	0.9307	30.69	0.8504	29.35	0.8127	29.15	0.8725	34.59	0.9511
Bicubic	$\times 4$	-	28.42	0.8104	26.00	0.7027	25.96	0.6675	23.14	0.6577	24.89	0.7866
EDSR [31]	$\times 4$	DIV2K	32.46	0.8968	28.80	0.7876	27.71	0.7420	26.64	0.8033	31.02	0.9148
DPDNN [32]	$\times 4$	DIV2K	31.72	0.8890	28.28	0.7730	27.44	0.7290	25.53	0.7680	-	-
RDN [33]	$\times 4$	DIV2K	32.47	0.8990	28.81	0.7871	27.72	0.7419	26.61	0.8028	31.00	0.9151
RNAN [34]	$\times 4$	DIV2K	32.49	0.8982	28.83	0.7878	27.72	0.7421	26.61	0.8023	31.09	0.9149
MoG-DUN [35]	$\times 4$	DIV2K	32.60	0.8998	28.84	0.7873	27.70	0.7403	26.63	0.8016	31.16	0.9169
RCAN [36]	$\times 4$	DIV2K	<u>32.63</u>	0.9002	28.87	0.7889	27.77	0.7436	<u>26.82</u>	<u>0.8087</u>	31.22	0.9173
SwinIR [46]	$\times 4$	DIV2K	32.72	0.9021	28.94	0.7914	27.83	0.7458	27.07	0.8164	31.67	0.9226
DGSM-UNet [10]	$\times 4$	DIV2K	32.48	0.8956	28.86	0.7864	27.71	0.7418	26.53	0.7986	31.00	0.9151
DGSM-Swin	$\times 4$	DIV2K	32.61	0.9005	28.91	0.7903	27.78	0.7445	26.73	0.8068	31.25	0.9193
DBPN [93]	$\times 4$	DIV2K+Flickr2K	32.47	0.8980	28.82	0.7860	27.72	0.7400	26.38	0.7946	30.91	0.9137
SwinIR [46]	$\times 4$	DIV2K+Flickr2K	32.92	0.9044	29.09	0.7950	27.92	0.7489	27.45	0.8254	32.03	0.9260
DGSM-Swin	$\times 4$	DIV2K+Flickr2K	<u>32.70</u>	0.9014	28.97	0.7917	27.83	0.7452	27.06	0.8142	31.58	0.9216

The bold entities denote the best performance.

TABLE III

ABLATION STUDY ON THE EFFECTS OF THE w -GENERATOR

Network	w/o the w -generator		with the w -generator	
	PSNR	SSIM	PSNR	SSIM
DGSM-UNet [10]	31.83	0.903	32.63	0.917
DGSM-Swin	35.93	0.951	36.55	0.959

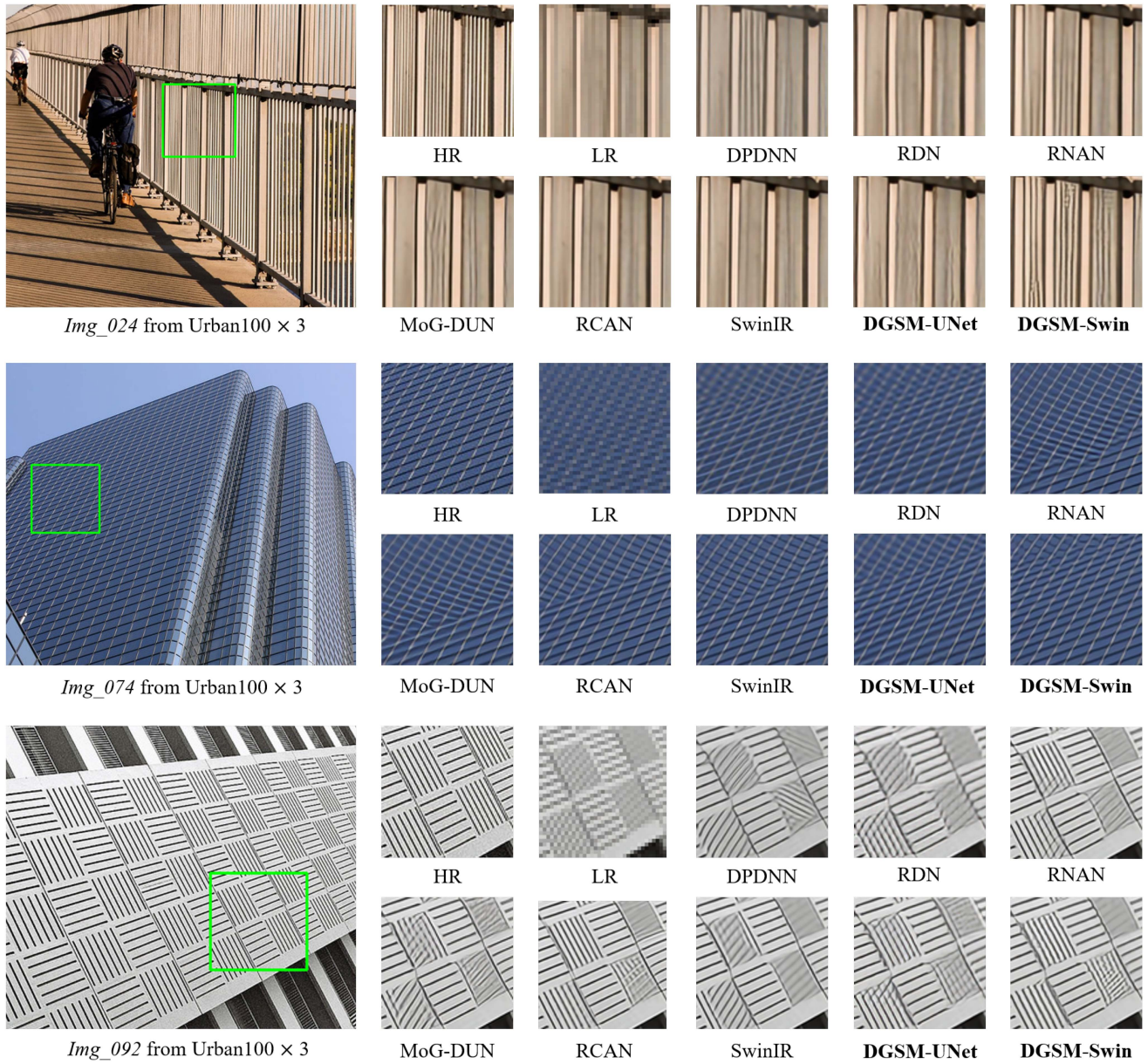
E. Complexity Analysis

In addition to the quantitative and qualitative results, we have also compared the proposed method with other competing SCI and SISR methods in terms of computational complexity (measured by the number of network parameters and FLOPS). Tables V and VI show the computational complexity comparisons of the competing SCI methods and the competing SISR methods, respectively. The FLOPS of the SCI methods and the SISR methods are evaluated on the input of sizes $256 \times 310 \times 1$ and $64 \times 64 \times 3$, respectively. For SCI, we have also implemented several variants of the proposed method DGSM-Swin, which have different numbers of stages T and STLs P . From Table V, we can observe that our methods

have fewer parameters and better results than λ -net [37] and TSA-Net [20]. Due to the multi-stage mechanism and the spatial-wise Swin Transformer, our proposed method has more parameters and FLOPS than HDNet [44] and MST [47]. However, our proposed network DGSM-Swin outperforms HDNet [44] and MST [47] by 2.21 dB and 1.37 dB in average PSNR, respectively. For SISR, from Table VI, our proposed method has fewer parameters and FLOPS and achieves results comparable with most competing methods. The two deep unfolding methods DPDNN [32] and MoG-DUN [35] have 3.2 and 2.6 times as many FLOPS as our proposed method DGSM-Swin, respectively. Compared with the best method SwinIR [46], we can save 2 M parameters and nearly half of FLOPS and achieve comparable results.

V. REAL DATA RESULTS

To further evaluate the effectiveness of the proposed method, we also conduct extensive real data experiments on SCI and SISR tasks. Some samples used for training and testing are shown in Fig. 6.

Fig. 8. Visual quality comparisons of different SISR methods for three sample images in the *Urban100* dataset (bicubic-downsampling, $\times 3$).TABLE IV
ABLATION STUDY ON THE EFFECTS OF THE NUMBER OF STAGES AND SWIN TRANSFORMER LAYERS (STLs) FOR SCI

Model	DGSM-Swin (gradient descent)										DGSM-Swin (exact solution)				
The number of stages (T)	1	1	1	2	2	2	3	3	3	4	4	2	2	3	4
The number of STLs (P)	2	4	6	2	4	6	2	4	6	2	4	2	4	2	2
PSNR	35.27	35.92	36.12	36.10	36.55	36.61	36.13	36.54	36.61	36.47	36.52	35.88	36.22	35.80	36.23
SSIM	0.947	0.953	0.956	0.955	0.959	0.960	0.957	0.960	0.961	0.959	0.961	0.945	0.948	0.945	0.951

The bold entities denote the best performance.

TABLE V
PARAMETERS, FLOPS, AND PERFORMANCE OF THE COMPETING SCI METHODS

Method	λ -net [37]	TSA-Net [20]	HDNet [44]	GAP-CCoT [23]	MST [47]	DGSM-UNet [10]	DGSM-Swin ($T=1, P=2$)	DGSM-Swin ($T=1, P=4$)	DGSM-Swin ($T=1, P=6$)	DGSM-Swin ($T=2, P=2$)	DGSM-Swin ($T=2, P=4$)
Params. (M)	62.64	44.25	2.37	8.04	2.03	3.76	6.79	9.37	11.96	6.80	9.38
FLOPS (G)	117.98	110.06	154.76	95.52	28.15	646.65	246.54	388.49	530.44	493.08	776.98
PSNR	28.53	31.46	34.34	35.16	35.18	32.63	35.28	35.92	36.12	36.10	36.55
SSIM	0.841	0.894	0.957	0.947	0.948	0.917	0.947	0.953	0.956	0.955	0.959

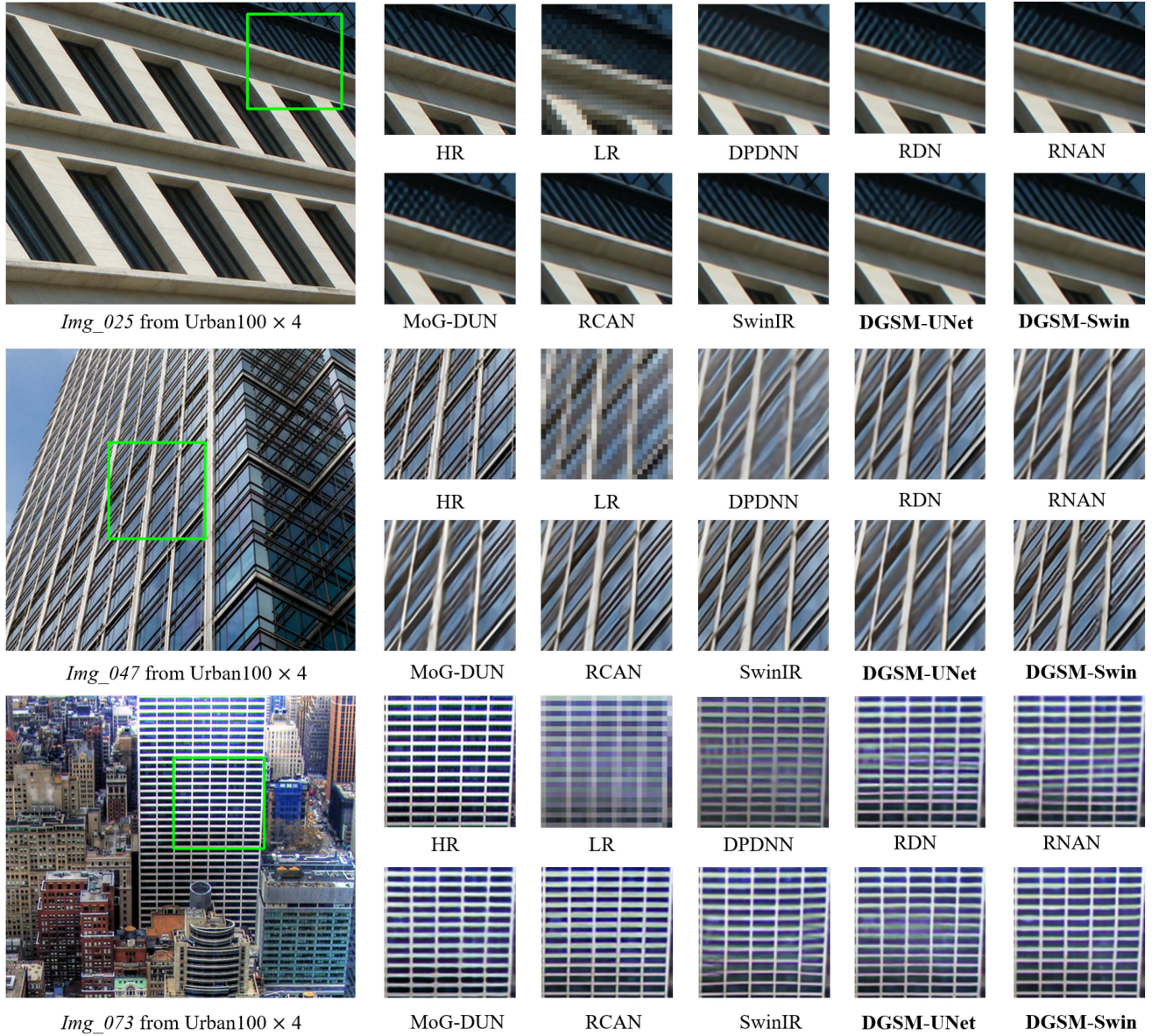


Fig. 9. Visual quality comparisons of different SISR methods for three sample images in the *Urban100* dataset (bicubic-downsampling, $\times 4$).

TABLE VI
PARAMETERS, FLOPS, AND PERFORMANCE OF THE COMPETING SISR METHODS (BICUBIC-DOWNSAMPLING, $\times 2$)

Method	EDSR [31]	DPDNN [32]	RDN [33]	RNAN [34]	MoG-DUN [35]	RCAN [36]	SwinIR [46]	DGSM-UNet [10]	DGSM-Swin
Params. (M)	40.73	1.36	22.12	9.11	5.75	15.44	11.75	1.72	9.75
FLOPS (G)	166.9	113.1	90.6	31.7	96.4	62.9	67.5	22.9	35.3
PSNR	38.11	37.75	28.24	38.17	38.25	38.27	38.35	38.08	38.24
SSIM	0.9602	0.9600	0.9614	0.9611	0.9614	0.9614	0.9620	0.9610	0.9615

A. Results on Real-World Spectral Compressive Imaging

We now conduct experiments on the real single disperser (SD) CASSI system [20] to verify the robustness of the proposed methods DGSM-UNet and DGSM-Swin. The real SD-CASSI system [20] can capture the real scene of spatial size 660×660 with 28 spectral bands ranging from 450 nm to 650 nm. Due to 54-pixel dispersion in the column dimension, the captured measurements have a spatial size of 660×714 . Similar to

TSA-Net [20], all scenes of the CAVE dataset and the KAIST dataset are used for training and 11-bit shot noise is injected to simulate the real measurements during training. We re-train the proposed method and compare the proposed method with λ -net [37], TSA-Net [20], HDNet [44], and MST [47]. Visual comparisons of the competing methods are shown in Fig. 10. The proposed method can recover more pleasant results and less blurring artifacts. Compared with DGSM-UNet, DGSM-Swin

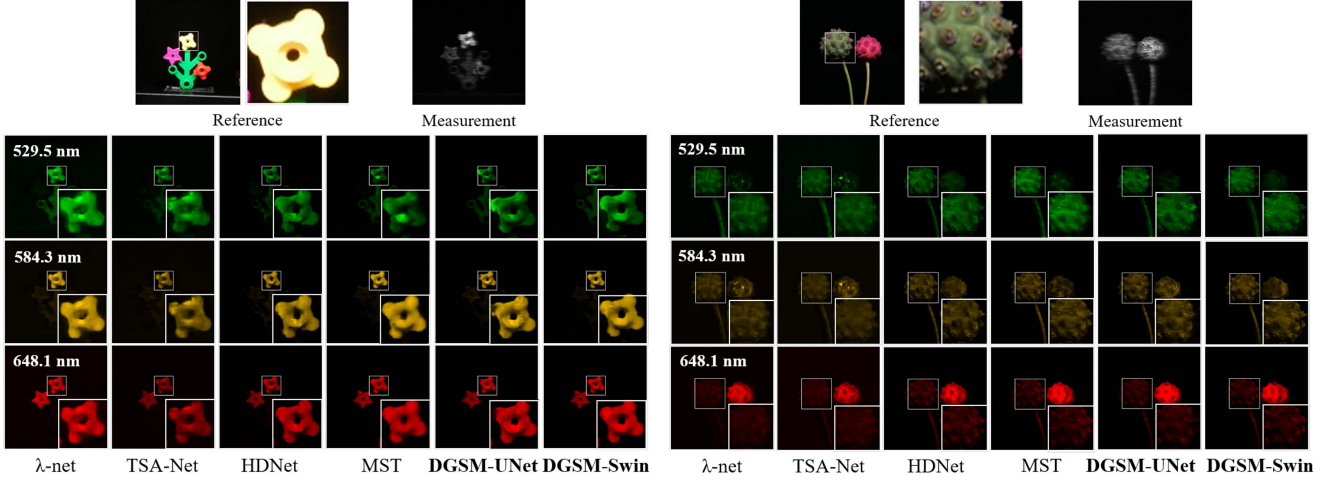


Fig. 10. Reconstructed images of two real scenes (*Scene 1* and *Scene 4*) with 3 out of 28 spectral channels by the competing methods. Zoom in for better view.

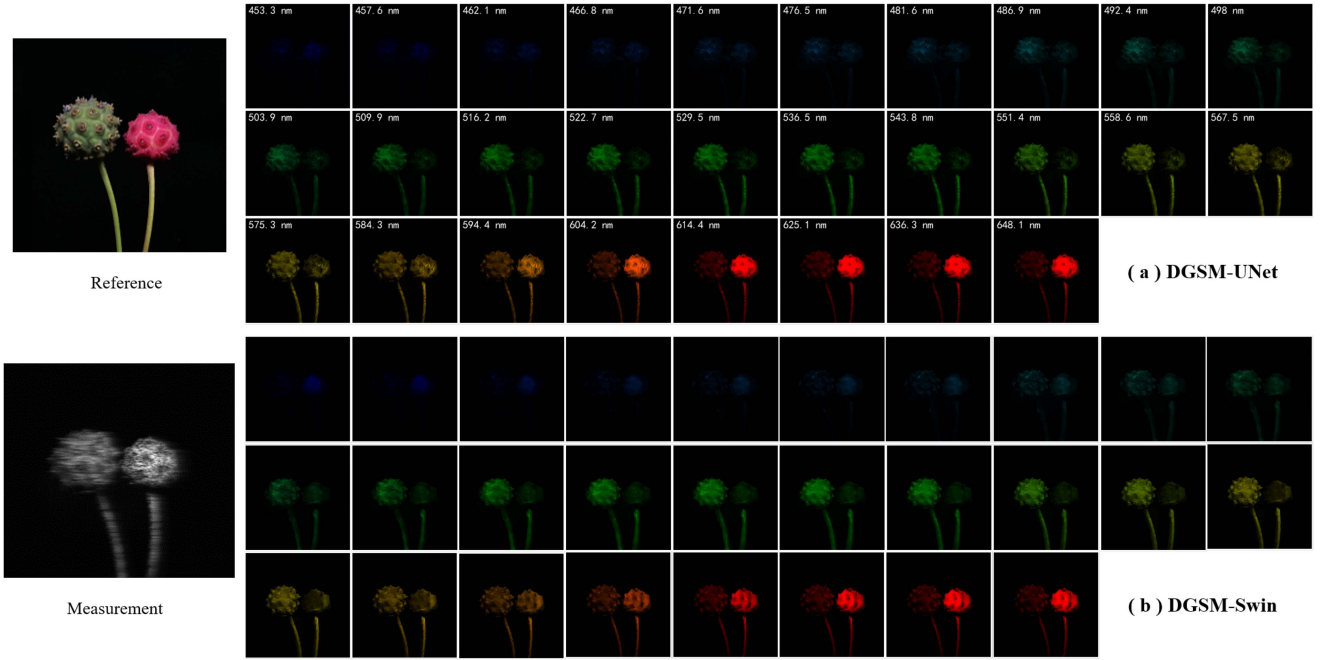


Fig. 11. Reconstructed images of the real scene (*Scene 4*) with 28 spectral channels by the proposed methods (a) *DGSM-UNet* [10] and (b) *DGSM-Swin*. Zoom in for better view.

can reconstruct more details of the textures due to the better w . Fig. 11 shows reconstructed images of the real scene (*Scene 4*) with 28 spectral bands by the proposed method. The visual results also reveal that the proposed methods have good model generalization ability.

B. Results on Real-World Single Image Super-Resolution

We further apply the proposed methods *DGSM-UNet* and *DGSM-Swin* to real-world SISR. We re-trained the proposed methods on the same training dataset with SwinIR [46] and used the degradation model proposed by BSRGAN [94] to simulate the LR images. We compare the proposed methods

with ESRGAN [95] and four state-of-the-art real-world image SR models RealSR [96], BSRGAN [94], Real-ESRGAN [97], and SwinIR [46]. All competing methods are tested on the RealSRSet [94] and the visual comparison results with the scaling factor $\times 4$ are shown in Fig. 12. From Fig. 12, we can see that our proposed methods can generate more pleasant results and suppress more noise and blurring artifacts.

VI. CONCLUSION

We have proposed an image reconstruction method *DGSM* for image reconstruction. Unlike existing works, our proposed

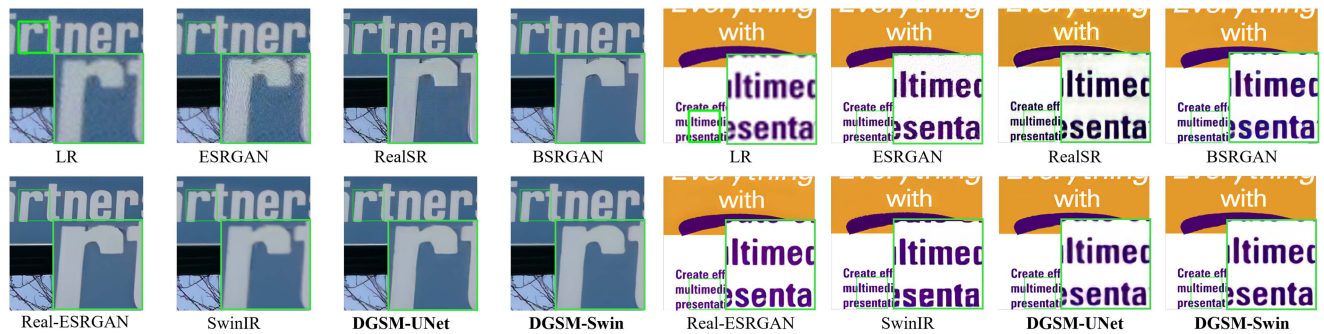


Fig. 12. Visual comparison of real-world image SR ($\times 4$) methods on real-world images. Zoom in for better view.

method is inspired by the Gaussian scale mixture prior. Specifically, the desired high-quality images were characterized by the GSM models, and then the reconstruction problem was formulated as an MAP estimation problem. Instead of only estimating the image means as in existing unfolding methods, we learn the means and the variances of the GSM models by a U-Net and construct an unfolding network DGSM-UNet. To learn long-range dependencies of images, we further proposed an enhanced variant DGSM-Swin to learn the means and the variances by a Transformer-based GSM prior network. Extensive experimental results show that DGSM-Swin outperforms DGSM-UNet by a large margin on SCI and SISR. Extensive experimental results on both synthetic and real datasets demonstrate that the proposed method DGSM outperforms existing state-of-the-art SCI and SISR algorithms.

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